

length

Length of largest array dimension

Syntax

```
L = length(X)
```

Description

`L = length(X)` returns the length of the largest array dimension in `X`. For vectors, the length is simply the number of elements. For arrays with more dimensions, the length is `max(size(X))`. The length of an empty array is zero. [example](#)

Examples

[collapse all](#)

Number of Vector Elements

Find the length of a uniformly spaced vector in the interval `[5,10]`.

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```
v = 5:10
```

[Get](#) ▼

```
v = 1×6
```

```
5    6    7    8    9   10
```

```
L = length(v)
```

[Get](#) ▼

```
L =  
6
```

Length of Rectangular Matrix

Find the length of a 3-by-7 matrix of zeros.

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```
X = zeros(3,7);  
L = length(X)
```

[Get](#) ▼

```
L =
7
```

String Array

Create a string array and compute its length, which is the number of elements in each row.

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```
X = ["a" "b" "c"; "d" "e" "f"]
```

[Get](#)

```
X = 2x3 string
    "a"    "b"    "c"
    "d"    "e"    "f"
```

```
L = length(X)
```

[Get](#)

```
L =
3
```

Length of Structure Fields

Create a structure with fields for Day and Month. Use the structfun function to apply length to each field.

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```
S = struct('Day',[1 13 14 26],'Month',{'Jan','Feb','Mar'})
```

[Get](#)

```
S = struct with fields:
    Day: [1 13 14 26]
    Month: {'Jan' 'Feb' 'Mar'}
```

```
L = structfun(@(field) length(field),S)
```

[Get](#)

```
L = 2x1

    4
    3
```

Input Arguments

X — Input array

scalar | vector | matrix | multidimensional array

Input array, specified as a scalar, vector, matrix, or multidimensional array.

Complex Number Support: Yes

Tips

- To find the number of characters in a string or character vector, use the [strlength](#) function.
- `length` does not operate on tables. To examine the dimensions of a table, use the `height`, `width`, or `size` functions.

Extended Capabilities

[expand all](#)**Tall Arrays**

Calculate with arrays that have more rows than fit in memory.

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Version History

Introduced before R2006a

See Also

[strlength](#) | [ndims](#) | [numel](#) | [size](#)

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